VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Machine Learning (23CS6PCMAL)

Submitted by

Deepthi M (1BM23CS088)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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(Affiliated To Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Machine Learning (23CS6PCMAL)" carried out by **Deepthi M (1BM23CS088),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of a Machine Learning (23CS6PCMAL) work prescribed for the said degree.

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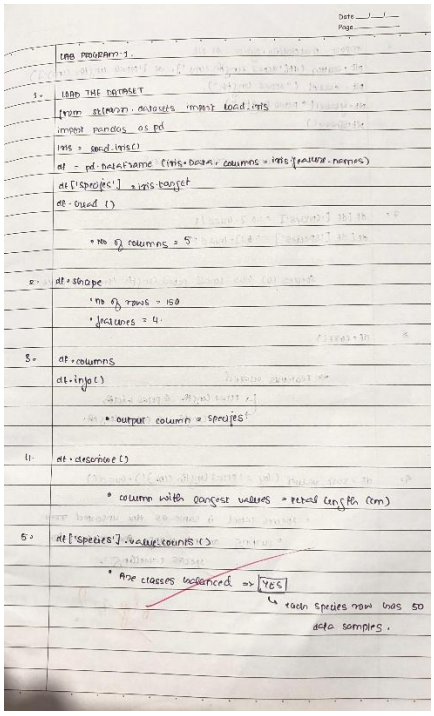
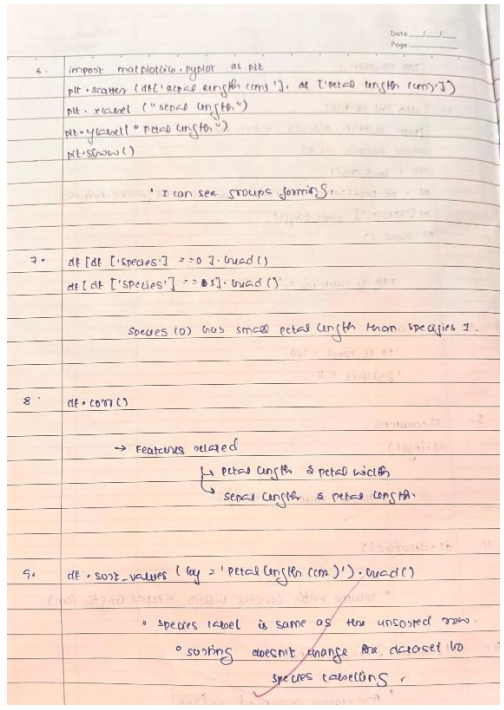
Github Link: https://github.com/Deepthi-1534/MACHINE-LEARNING-LAB

Program 1

Write a python program to import and export data using Pandas library functions

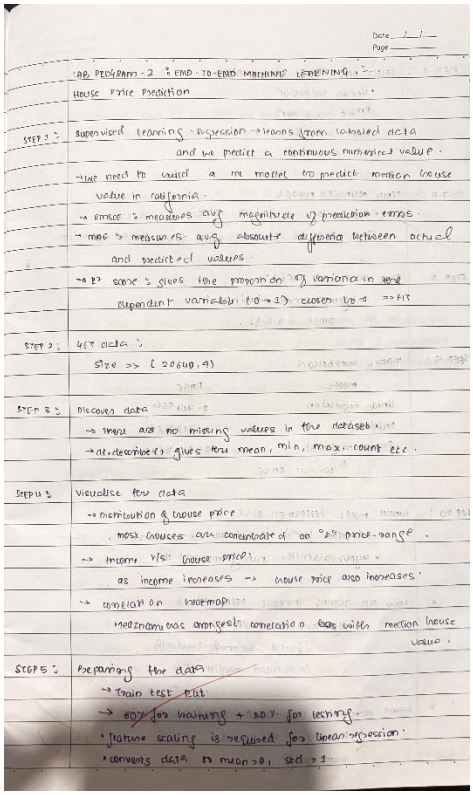


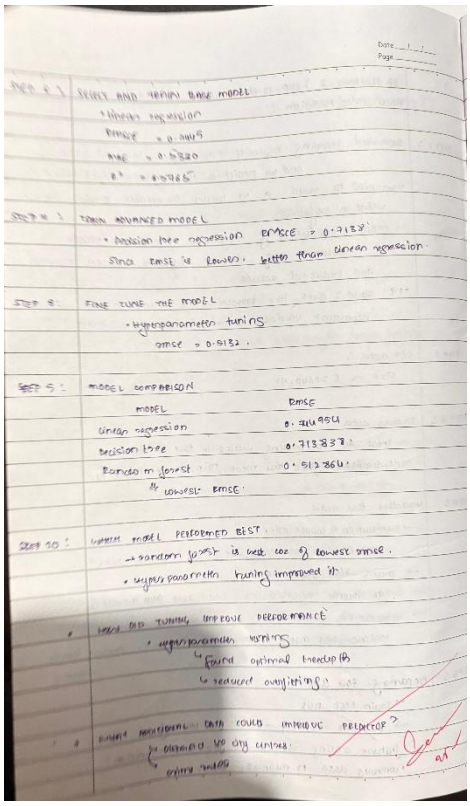



```
import pandas as pd
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv"
df = pd.read_csv(url)
print(df.head())
df.to_csv("output.csv", index=False)
df.to_excel("output.xlsx", index=False)
df.to_json("output.json", orient="records")
df.to_html("output.html", index=False)
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Program – 2: Demonstrate various data pre-processing techniques for a given dataset

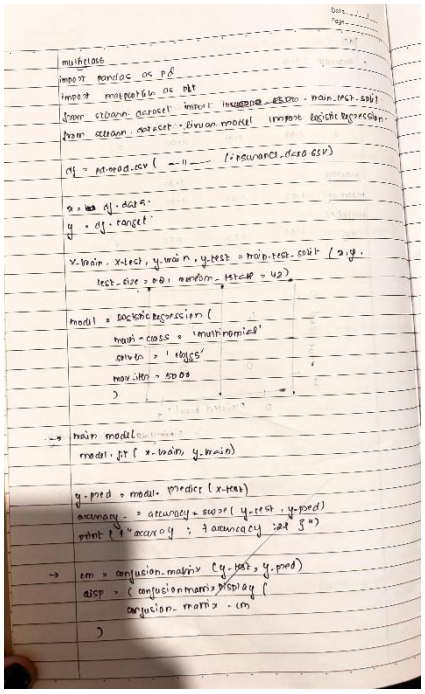


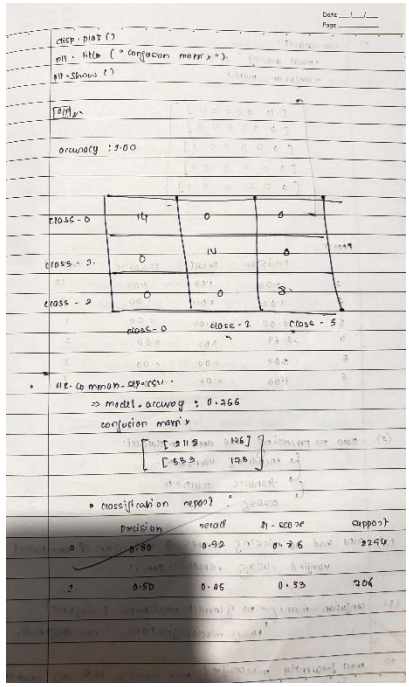


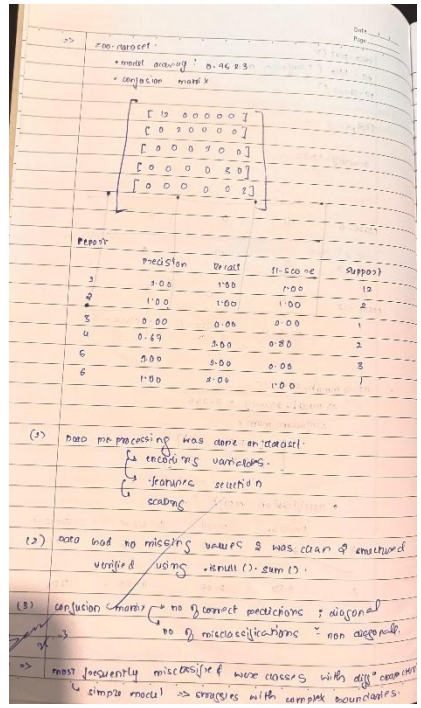

```
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, MinMaxScaler, LabelEncoder

url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv"
df = pd.read_csv(url)

print("Original Data:")
print(df.head())

df = df[['survived','pclass','sex','age','sibsp','parch','fare','embarked']]

df['age'].fillna(df['age'].median(), inplace=True)
df['embarked'].fillna(df['embarked'].mode()[0], inplace=True)

df.drop_duplicates(inplace=True)

le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex'])
```

```
df['embarked'] = le.fit_transform(df['embarked'])

X = df.drop('survived', axis=1)
y = df['survived']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler_standard = StandardScaler()
X_train_std = scaler_standard.fit_transform(X_train)
X_test_std = scaler_standard.transform(X_test)

scaler_minmax = MinMaxScaler()
X_train_mm = scaler_minmax.fit_transform(X_train)
X_test_mm = scaler_minmax.transform(X_test)

print("\nAfter Preprocessing:")
print("Shape:", X_train.shape)
print("Standardized Sample:\n", X_train_std[:5])
print("MinMax Scaled Sample:\n", X_train_mm[:5])
```

Program-3 : Implement Linear and Multi-Linear Regression algorithm using appropriate dataset

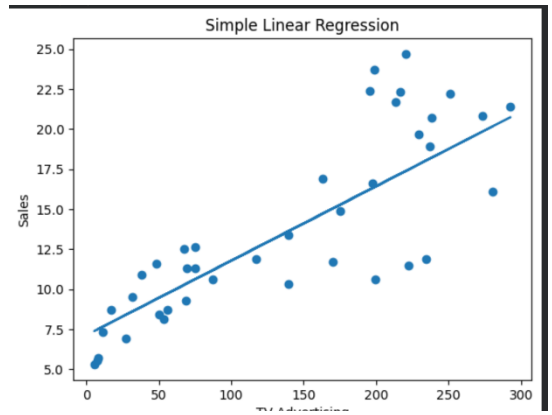






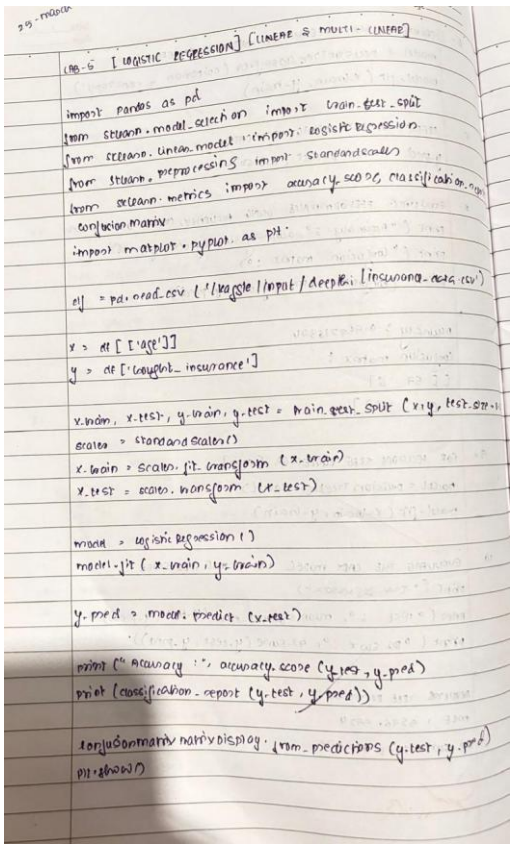
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

url = "https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv"
df = pd.read_csv(url)

X = df[['TV']]
y = df['sales']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

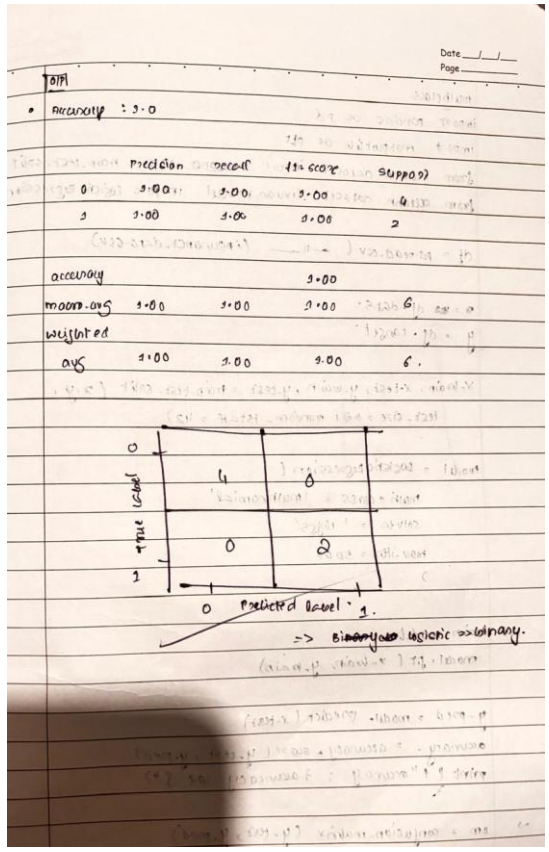
```
print(mean_squared_error(y_test, y_pred))
print(r2_score(y_test, y_pred))

plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("TV Advertising")
plt.ylabel("Sales")
plt.title("Simple Linear Regression")
plt.show()
```




```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

url = "https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv"
df = pd.read_csv(url)

X = df[['TV', 'radio', 'newspaper']]
y = df['sales']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print(mean_squared_error(y_test, y_pred))
print(r2_score(y_test, y_pred))
3.1740973539761033
0.899438024100912
```

Program – 4 : Build Logistic Regression Model for a given dataset



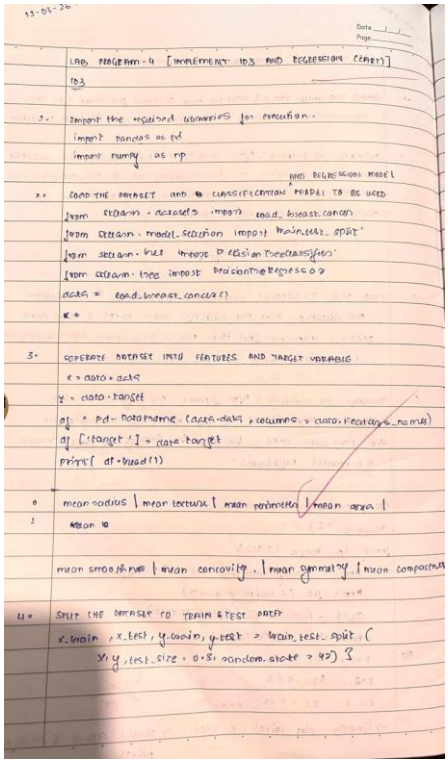


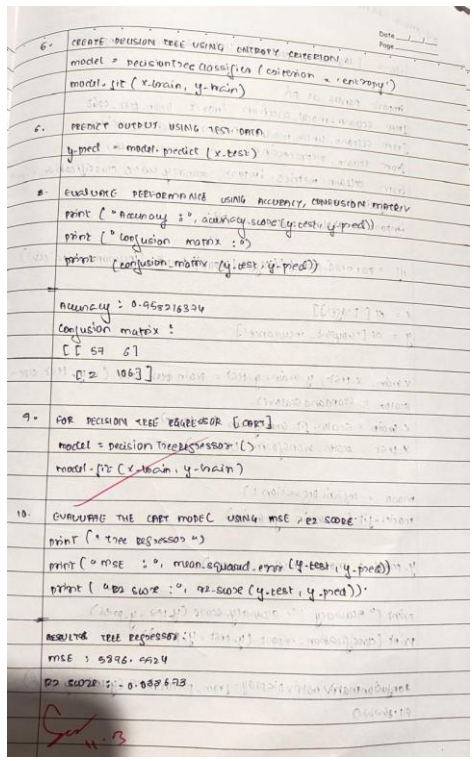

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.csv"

columns = [
    "pregnancies", "glucose", "blood_pressure", "skin_thickness",
    "insulin", "bmi", "diabetes_pedigree", "age", "outcome"
]

df = pd.read_csv(url, names=columns)
```

```
X = df.drop("outcome", axis=1)
y = df["outcome"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print(accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))

y_prob = model.predict_proba(X_test)[:, 1]

plt.hist(y_prob, bins=10)
plt.xlabel("Predicted Probability")
plt.ylabel("Frequency")
plt.title("Prediction Probability Distribution")
plt.show()
```

```
Output : 0.7532467532467533
[[79 20]
 [18 37]]
              precision    recall  f1-score   support

           0       0.81      0.80      0.81        99
           1       0.65      0.67      0.66        55

    accuracy                           0.75       154
   macro avg       0.73      0.74      0.73       154
weighted avg       0.76      0.75      0.75       154
```

Program-5 : Use an appropriate data set for building the decision tree (ID3) and apply this knowledge to classify a new sample.





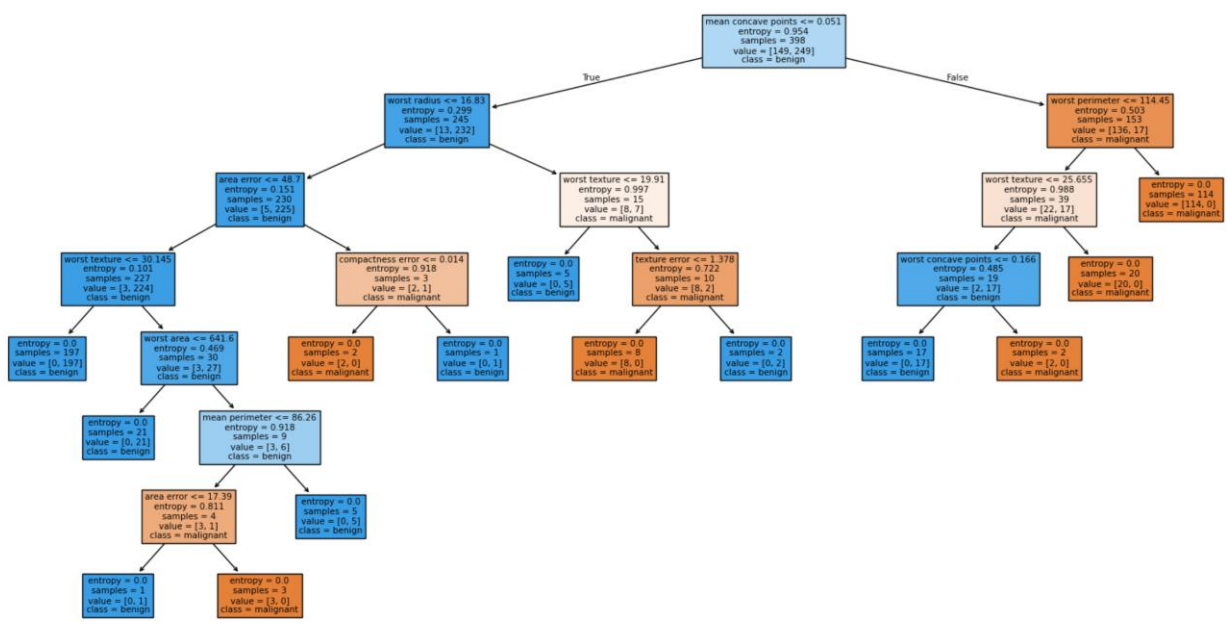
```
import pandas as pd
import numpy as np

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
data = load_breast_cancer()

X = data.data
y = data.target
df = pd.DataFrame(data.data, columns=data.feature_names)
df['target'] = data.target

print(df.head())
```

```
mean radius  mean texture  mean perimeter  mean area  mean smoothness  \
0       17.99        10.38          122.80     1001.0         0.11840
1       20.57        17.77          132.90     1326.0         0.08474
2       19.69        21.25          130.00     1203.0         0.10960
3       11.42        20.38           77.58      386.1         0.14250
4       20.29        14.34          135.10     1297.0         0.10030

   mean compactness  mean concavity  mean concave points  mean symmetry  \
0           0.27760          0.3001              0.14710         0.2419
1           0.07864          0.0869              0.07017         0.1812
2           0.15990          0.1974              0.12790         0.2069
3           0.28390          0.2414              0.10520         0.2597
4           0.13280          0.1980              0.10430         0.1809

   mean fractal dimension  ...  worst texture  worst perimeter  worst area  \
0                 0.07871  ...          17.33           184.60      2019.0
1                 0.05667  ...          23.41           158.80      1956.0
2                 0.05999  ...          25.53           152.50      1709.0
3                 0.09744  ...          26.50            98.87       567.7
4                 0.05883  ...          16.67           152.20      1575.0
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
model = DecisionTreeClassifier(criterion='entropy')

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
Accuracy: 0.9532163742690059
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

Confusion Matrix:
[[ 57   6]
 [  2 106]]

print("Classification Report:")
print(classification_report(y_test, y_pred))
Classification Report:
```

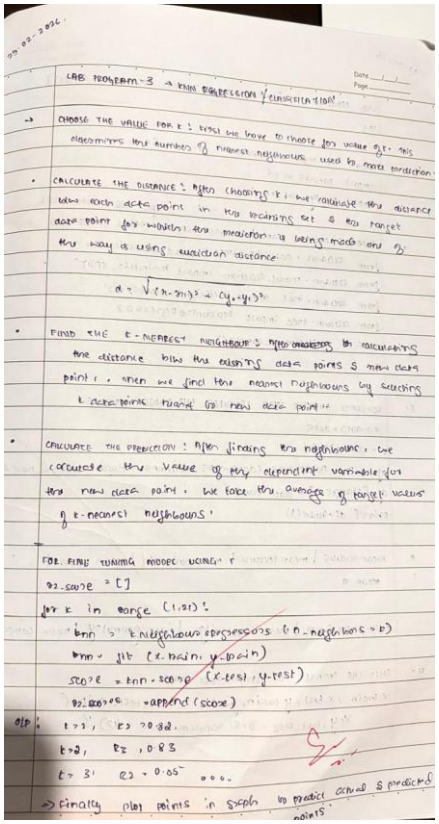
	precision	recall	f1-score	support
0	0.97	0.90	0.93	63
1	0.95	0.98	0.96	108
accuracy			0.95	171
macro avg	0.96	0.94	0.95	171
weighted avg	0.95	0.95	0.95	171

```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(20,10))

plot_tree(
    model,
    feature_names=data.feature_names,
    class_names=data.target_names,
    filled=True
)

plt.show()
```



```
import pandas as pd
import numpy as np

from sklearn.datasets import load_diabetes
from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor

from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
data = load_diabetes()
_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
model = DecisionTreeRegressor()

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix")
print(confusion_matrix(y_test, y_pred))

print("Classification Report")
print(classification_report(y_test, y_pred))
Accuracy: 0.8333333333333334
Confusion Matrix
[[19  0  0]
 [ 2 18  1]
 [ 2  4  8]]
Classification Report
              precision    recall  f1-score   support

           0       0.83      1.00      0.90        19
           1       0.82      0.86      0.84        21
           2       0.89      0.57      0.70        14

    accuracy                           0.83        54
   macro avg       0.84      0.81      0.81        54
weighted avg       0.84      0.83      0.82        54
```

Program -6 : Build KNN Classification model for a given dataset.



LAB PROGRAM-3 → KNN REGRESSION / CLASSIFICATION

→ CHOOSE THE VALUE FOR K : First we have to choose for value of k. This determines the number of nearest neighbours used to make prediction.

• CALCULATE THE DISTANCE : After choosing k, we calculate the distance b/w each data point in the training set & the target data point for which the prediction is being made. One of the way is using euclidean distance.

$$d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$

• FIND THE K-NEAREST NEIGHBOUR : After calculating the distance b/w the existing data points & new data point, we find the nearest neighbours by selecting k data points nearest to new data point.

• CALCULATE THE PREDICTION : After finding the neighbours, we calculate the value of the dependent variable for the new data point. We take the average of target values of k-nearest neighbours.

FOR FINE TUNING MODEL USING K

```
R2_score = []
for k in range (1,21):
    knn = KNeighborsRegressor (n_neighbors = k)
    knn.fit (x_train, y_train)
    score = knn.score (x_test, y_test)
    R2_score.append (score)
```

o/p : k=1, R2 = 0.82.
k=2, R2 = 0.83
k=3, R2 = 0.05 ....

→ Finally plot points in graph to predict actual & predicted points.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/breast-cancer.csv"

columns = [
    "age", "menopause", "tumor_size", "inv_nodes", "node_caps",
    "deg_malig", "breast", "breast_quad", "irradiat", "class"
]

df = pd.read_csv(url, names=columns)

print("Dataset Preview:")
print(df.head())
```

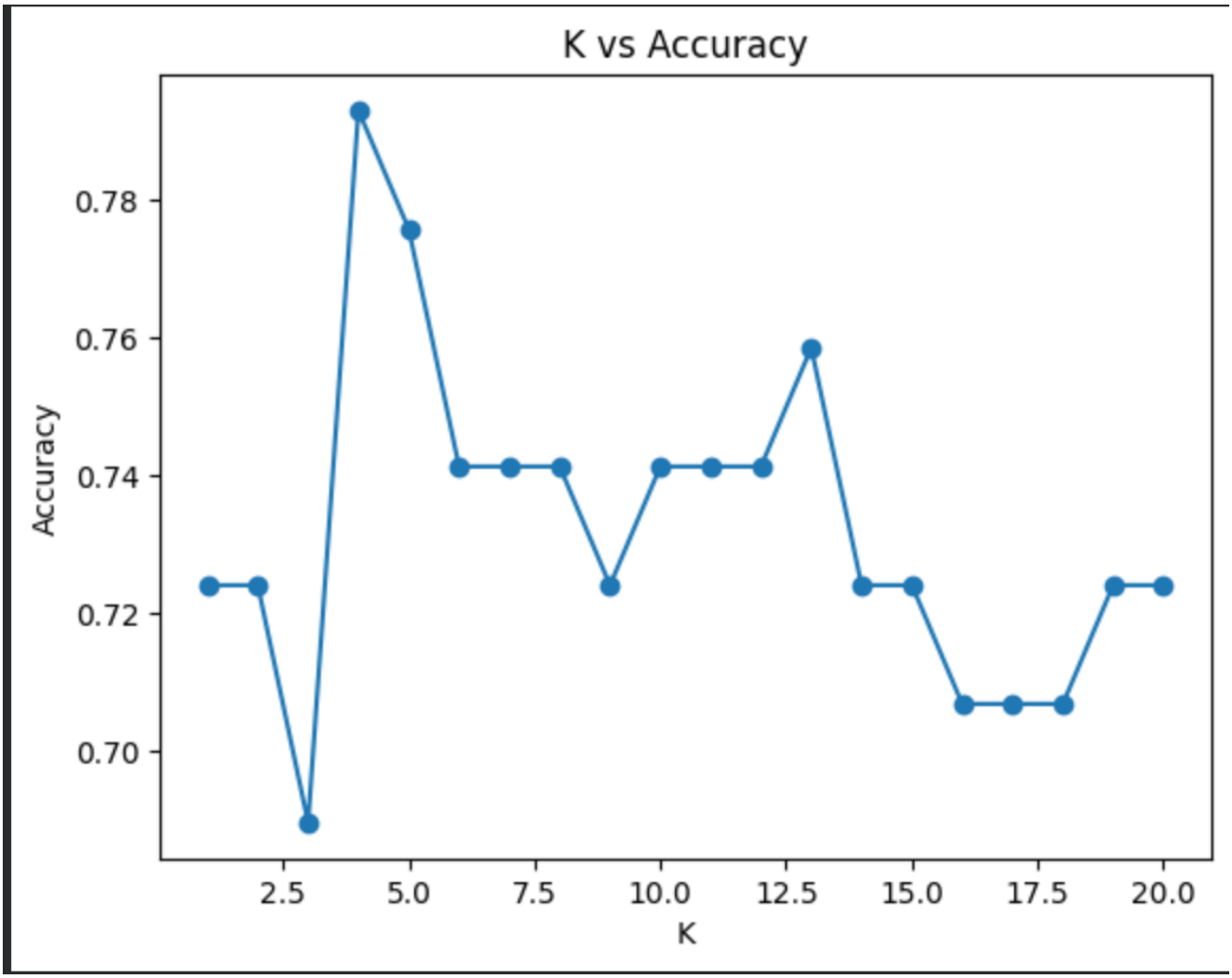
```
X = df.drop("class", axis=1)
y = df["class"]

X = pd.get_dummies(X)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nAccuracy:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

accuracy_scores = []

for k in range(1, 21):
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train, y_train)
    preds = knn.predict(X_test)
    accuracy_scores.append(accuracy_score(y_test, preds))

# Plot
plt.plot(range(1, 21), accuracy_scores, marker='o')
plt.xlabel("K")
plt.ylabel("Accuracy")
plt.title("K vs Accuracy")
plt.show()
```

Accuracy: 0.7758620689655172

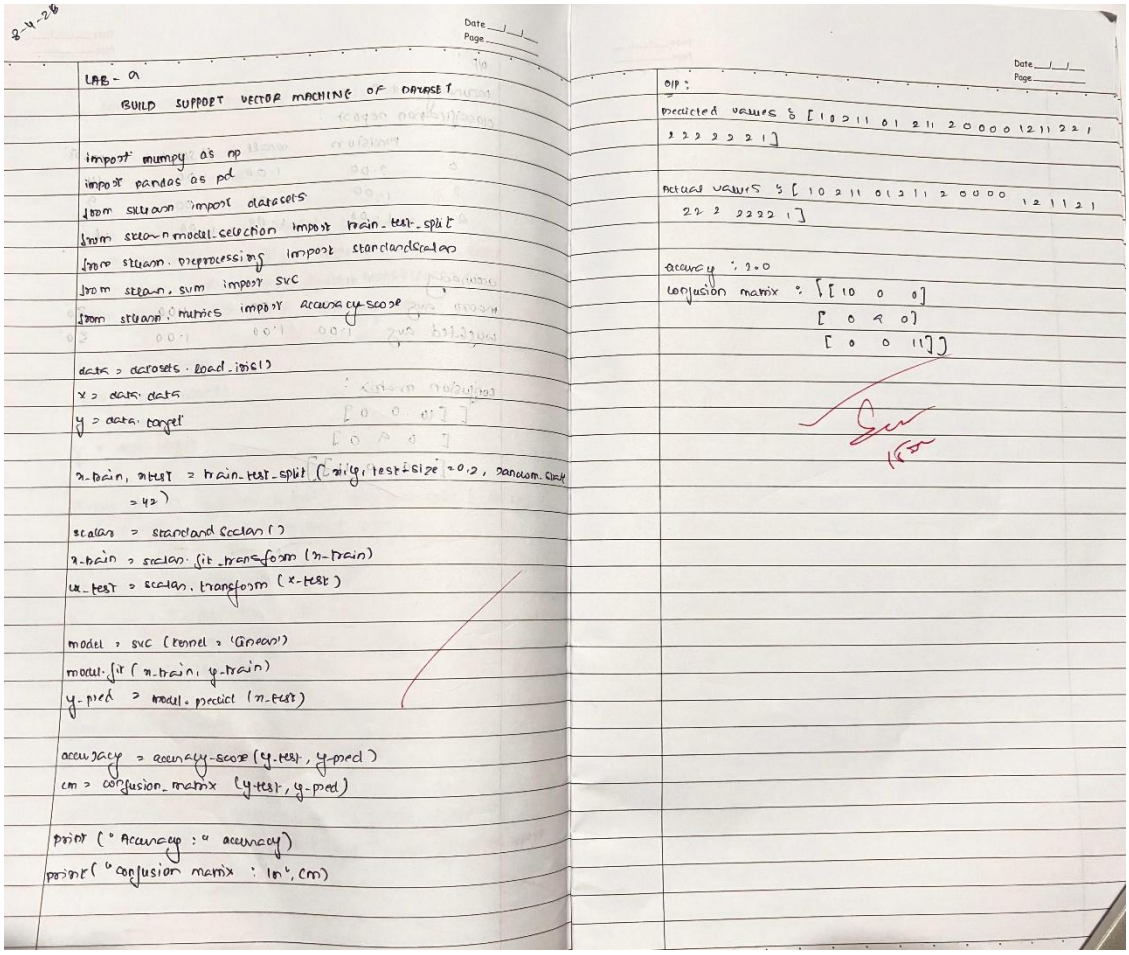
Confusion Matrix:
[[39 1]
 [12 6]]

Classification Report:

	precision	recall	f1-score	support
'no-recurrence-events'	0.76	0.97	0.86	40
'recurrence-events'	0.86	0.33	0.48	18
accuracy			0.78	58
macro avg	0.81	0.65	0.67	58
weighted avg	0.79	0.78	0.74	58



Program-7 : Build Support vector machine model for a given dataset



LAB - 9

BUILD SUPPORT VECTOR MACHINE OF DATASET

```
import numpy as np
import pandas as pd
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import standardScalar
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

data = datasets.load_iris()
X = data.data
y = data.target

X_train, X_test = train_test_split(X, y, test_size=0.2, random_state
=42)
scalar = standardScalar()
X_train = scalar.fit_transform(X_train)
X_test = scalar.transform(X_test)

model = SVC(kernel='linear')
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("Accuracy:", accuracy)
print("confusion matrix:\n", cm)
```

```
O/P :
predicted values : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 2 1
2 2 2 2 2 2 1]

Actual values : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 1
2 2 2 2 2 2 1]

accuracy : 1.0
confusion matrix : [[10 0 0]
                    [ 0 9 0]
                    [ 0 0 11]]
```

import numpy as np
import pandas as pd

from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
data = datasets.load_iris()

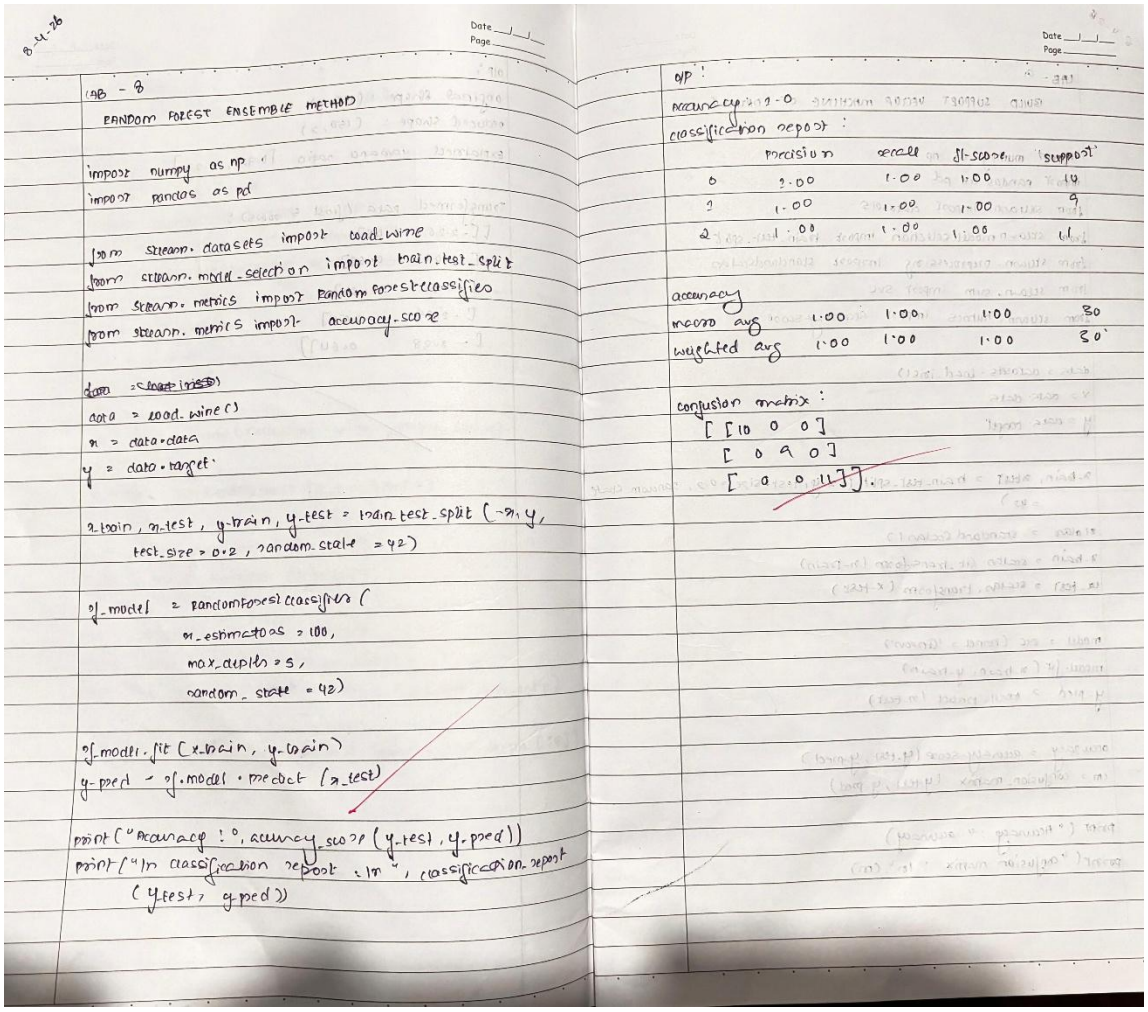
X = data.data
y = data.target

Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42

```
)

# Feature scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Model
model = SVC(kernel='linear')

# Train
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluation
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("Predicted values:", y_pred)
print("Actual values:", y_test)
print("Accuracy:", accuracy)
print("Confusion Matrix:\n", cm)
```

```
Predicted values: [1 0 2 1 1 0 1 2 2 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Actual values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy: 0.9666666666666667
Confusion Matrix:
 [[10  0  0]
 [ 0  8  1]
 [ 0  0 11]]
```

Program-8 : Implement Random forest ensemble method on a given dataset.

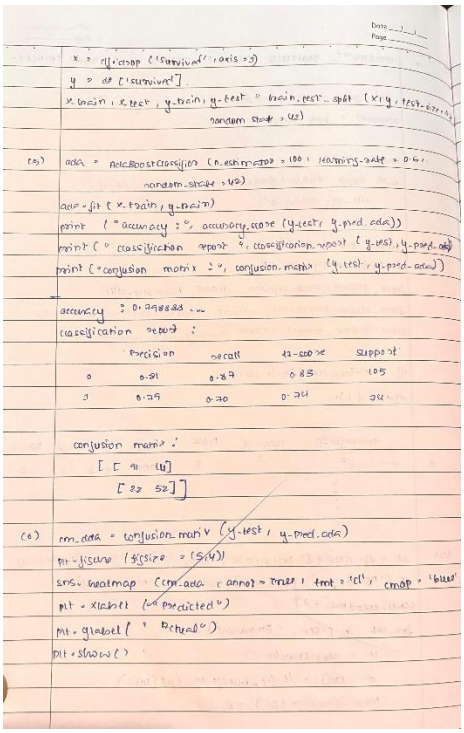


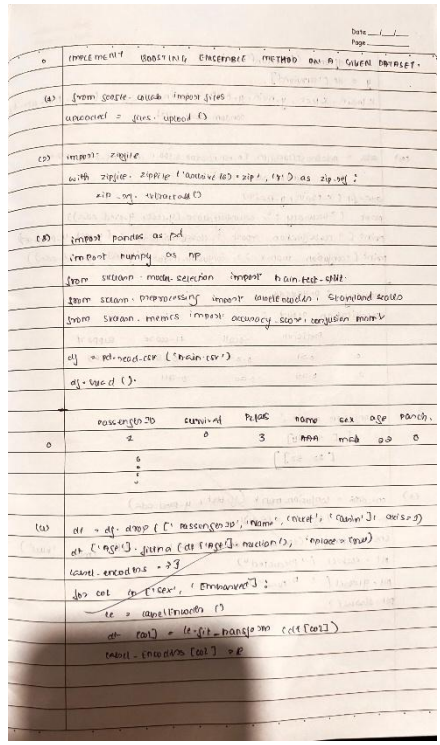
```
import numpy as np
import pandas as pd

from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Load dataset
data = load_wine()

X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
)

# Model
model = RandomForestClassifier(
    n_estimators=100,
    max_depth=5,
    random_state=42
)

# Train
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluation
accuracy = accuracy_score(y_test, y_pred)
report = classification_report(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("Accuracy:", accuracy)
print("\nClassification Report:\n", report)
```

```
•••  Accuracy: 1.0

     Classification Report:
                   precision    recall  f1-score   support

                0       1.00      1.00      1.00        14
                1       1.00      1.00      1.00        14
                2       1.00      1.00      1.00         8

            accuracy                           1.00        36
           macro avg       1.00      1.00      1.00        36
        weighted avg       1.00      1.00      1.00        36


     Confusion Matrix:
      [[14  0  0]
      [ 0 14  0]
      [ 0  0  8]]
```

Program-9 : Implement Boosting ensemble method on a given dataset.

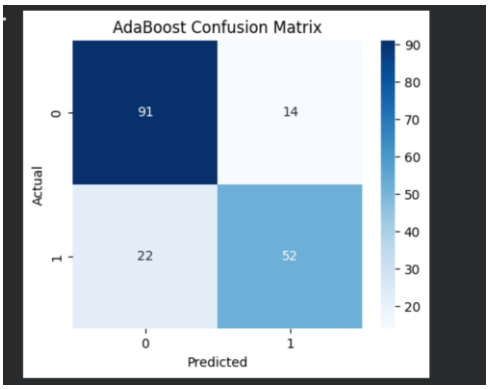




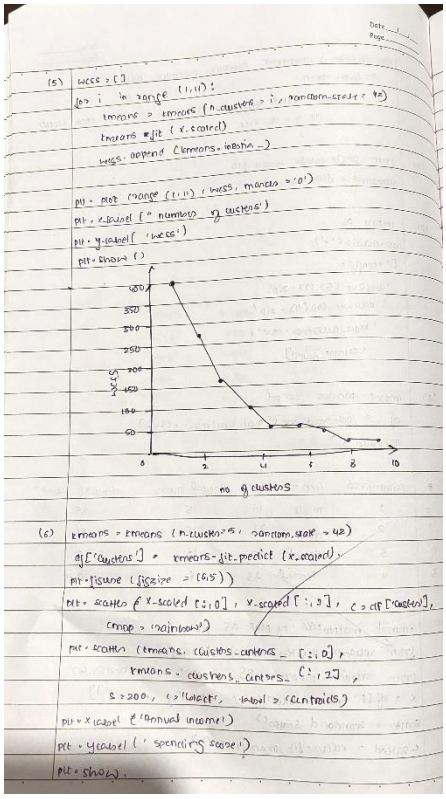
```
from google.colab import files
uploaded = files.upload()
import zipfile

# Extract ZIP to current directory
with zipfile.ZipFile('archive (6).zip', 'r') as zip_ref:
    zip_ref.extractall('.')
mport pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import AdaBoostClassifier, GradientBoostingClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Step 2: Load dataset
df = pd.read_csv('train.csv')
df = df.drop(['PassengerId', 'Name', 'Ticket', 'Cabin'], axis=1)

# Fill missing values
df['Age'].fillna(df['Age'].median(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

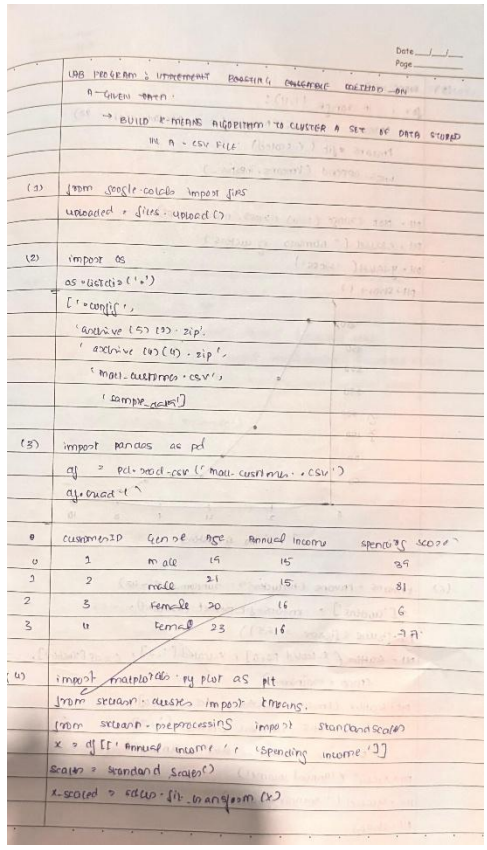
```
# Encode categorical variables
label_encoders = {}
for col in ['Sex', 'Embarked']:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    label_encoders[col] = le

# Split features and target
X = df.drop('Survived', axis=1)
y = df['Survived']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Scale features (optional but recommended for boosting)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
ada = AdaBoostClassifier(n_estimators=100, learning_rate=0.5, random_state=42)
ada.fit(X_train_scaled, y_train)
y_pred_ada = ada.predict(X_test_scaled)

# Evaluate AdaBoost
print("AdaBoost Accuracy:", accuracy_score(y_test, y_pred_ada))
print("AdaBoost Classification Report:\n", classification_report(y_test, y_pred_ada))
print("AdaBoost Confusion Matrix:\n", confusion_matrix(y_test, y_pred_ada))
```

```
AdaBoost Accuracy: 0.7988826815642458
AdaBoost Classification Report:
               precision    recall  f1-score   support

           0       0.81      0.87      0.83       105
           1       0.79      0.70      0.74        74

    accuracy                           0.80       179
   macro avg       0.80      0.78      0.79       179
weighted avg       0.80      0.80      0.80       179

AdaBoost Confusion Matrix:
 [[91 14]
 [22 52]]
```



Program-10: Build k-Means algorithm to cluster a set of data stored in a .CSV file.





from google.colab import files

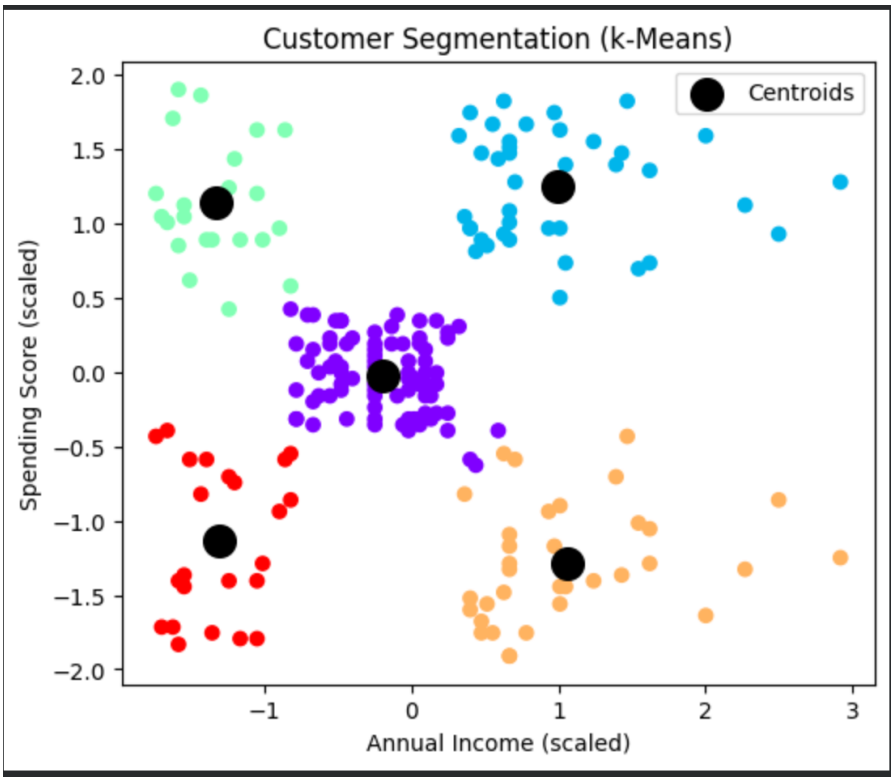
# Upload your ZIP file
uploaded = files.upload()
import pandas as pd

# Replace filename if your CSV has a different name
df = pd.read_csv('Mall_Customers.csv')

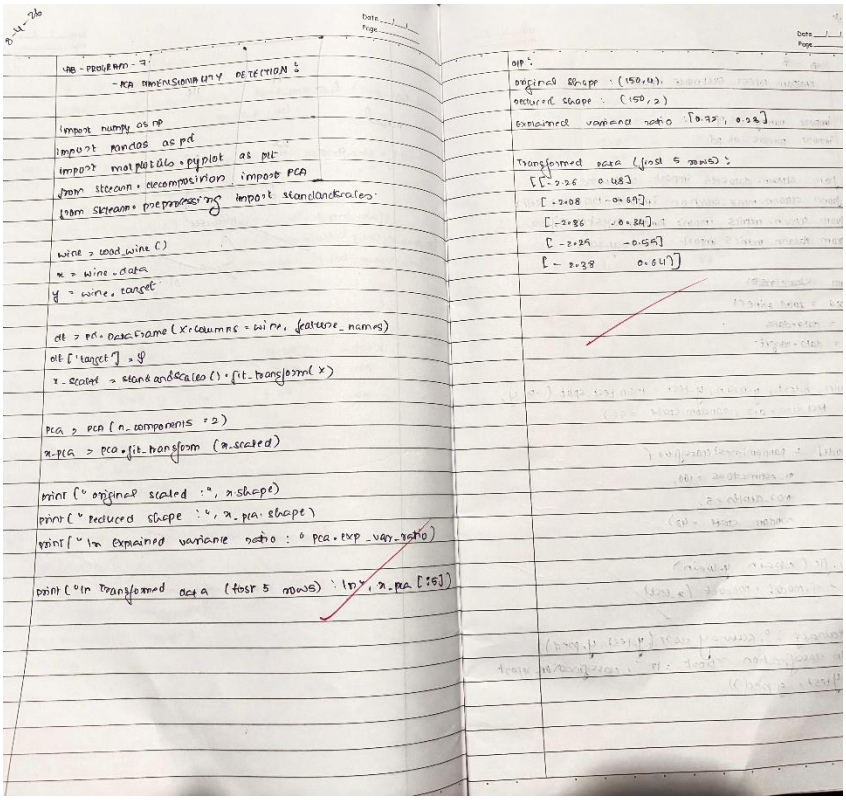
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

```
plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()

kmeans = KMeans(n_clusters=5, random_state=42)
df['Cluster'] = kmeans.fit_predict(X_scaled)
lt.figure(figsize=(6,5))
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=df['Cluster'], cmap='rainbow')
plt.scatter(kmeans.cluster_centers_[:, 0],
            kmeans.cluster_centers_[:, 1],
            s=200, c='black', label='Centroids')
plt.xlabel('Annual Income (scaled)')
plt.ylabel('Spending Score (scaled)')
plt.title('Customer Segmentation (k-Means)')
plt.legend()
plt.show()
```



Program-11 : Implement Dimensionality reduction using Principal Component Analysis (PCA) method.

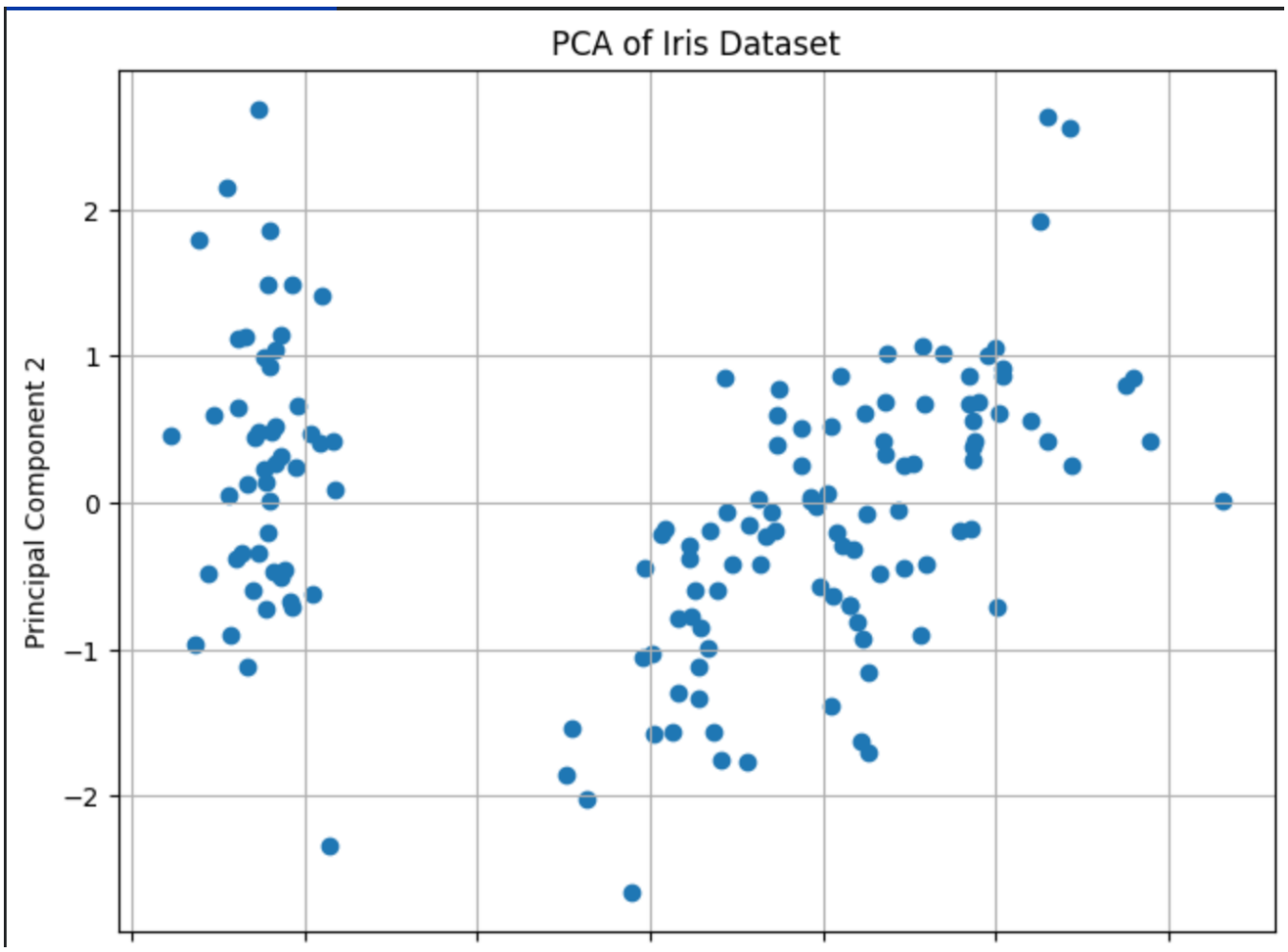


```
# Import libraries
import pandas as pd
import numpy as np
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

# Load built-in dataset
iris = load_iris()

# Convert to DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)

# Display first 5 rows
print("Original Dataset:")
print(df.head())

# Standardize the data
scaler = StandardScaler()
```

```
scaled_data = scaler.fit_transform(df)

# Apply PCA
pca = PCA(n_components=2)
principal_components = pca.fit_transform(scaled_data)

# Create new dataframe with PCA components
pca_df = pd.DataFrame(
    data=principal_components,
    columns=['Principal Component 1', 'Principal Component 2']
)

print("\nPCA Output:")
print(pca_df.head())

# Variance explained
print("\nExplained Variance Ratio:")
print(pca.explained_variance_ratio_)

# Plot graph
plt.figure(figsize=(8,6))
plt.scatter(
    pca_df['Principal Component 1'],
    pca_df['Principal Component 2']
)
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('PCA of Iris Dataset')
plt.grid(True)
plt.show()
```

```
Original Dataset:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2
```

PCA Output:

	Principal Component 1	Principal Component 2
0	-2.264703	0.480027
1	-2.080961	-0.674134
2	-2.364229	-0.341908
3	-2.299384	-0.597395
4	-2.389842	0.646835

Explained Variance Ratio:
[0.72962445 0.22850762]